

BT244: Microeconomics Notes

Steven DeFalco

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Contents

1	Principles of Economics	2
2	Economic Models: Trade-offs and Trade	2
2.1	The Circular-Flow Diagram	3
2.2	The Production Possibilities Frontier	3
2.3	Positive versus Normative Economics	3
3	Supply and Demand	4
3.1	Competitive Markets	4
3.2	Demand	4
3.3	Supply	5
4	Consumer and Producer Surplus	6
5	Price Controls and Quotas	8
6	Elasticity	9
6.1	Factors that Determine Price Elasticity of Demand	10
7	Taxes	11
8	Decision Making by Individuals and Firms	12
8.1	Behavioral Economics	13
9	The Rational Consumer	14
10	Behind the Supply Curve: Inputs and Costs	16

1 Principles of Economics

Economics is the study of how individuals make decision: how we produce, distribute, consume goods and services in a society where resources are scarce. The principles of economics are as follows:

- Choices are necessary because resources are scarce.
 - A **resource** is anything that can be used to produce something else.
 - A resource is **scarce** when there is not enough of the resource available to satisfy all the various ways a society wants to use it.
- The true cost of something is its **opportunity cost** (def: what you must give up in order to get something).
- *How much* is a decision at the margin (for each additional unit).
 - A **marginal decision** is a decision made at the margins of an activity about whether to do a bit more or a bit less of that activity
 - **Marginal analysis** is the study of margin decisions.
- People respond to incentives (def: anything that offers rewards to people who change their behavior), exploiting opportunities to make themselves better off.

Trade allows us to consume more than we otherwise could; gains from trade arise from specialization. **Specialization** is the situation in which each person specializes in the task that he or she is good at performing. The principles of the interaction of individual choices are as follows:

1. There are gains from trade.
2. Markets (def: a place where a good or service is exchanged) move toward equilibrium (def: an economic situation in which no individual would be better off doing something different)
3. Resources should be used efficiently to achieve society's goals. Economy is efficient if it takes all opportunities to make some people better off without making other people worse off.
4. Markets usually lead to efficiency (def: all the opportunities to make people better off have been exploited), but when they don't, government intervention can improve society's welfare.

2 Economic Models: Trade-offs and Trade

A **model** is a simplified representation of a real situation that is used to better understand real-life situations. By keeping our models simple, we can focus on the change of one variable, assuming everything else stays the same; this is the *other things equal* assumption (*ceteris paribus*).

2.1 The Circular-Flow Diagram

The *circular-flow diagram* represents the transactions in an economy by flows around a circle.

Definition 2.1 A *household* is a person or a group of people that share their income

Definition 2.2 A *firm* is an organization that produces goods and services for sale.

Firms sell goods and services that they produce to households in **markets for goods and services**. Firms buy the resources they need to produce goods and services in **factor markets**. Main factors of production are land, labor, physical capital, and human capital. An economy's **income distribution** is the way in which total income is divided among the owners of the various factors of production.

2.2 The Production Possibilities Frontier

The *production possibilities frontier (PPF)* is a diagram that shows the combinations of two goods that are possible for a society to produce at full employment.

An economy is **efficient** if there are no missed opportunities. An economy is **efficient in production** if there are no missed opportunities in production. It is efficient in production if it could not produce more of any one good without producing less of something else—if it's on the PPF. An economy is inefficient in production if it could produce more of some things without producing less of others. The economy is **efficient in allocation** if it allocates its resources so that consumers are as well off as possible. Efficiency requires both efficiency in production and efficiency in allocation.

Economic growth means an expansion of the economy's production possibilities. Economic growth can be caused by:

1. **An increase in factors of production:** resources used to produce goods and services (land, labor, physical capital, and human capital).
2. **Better technology:** the technical means for producing goods and services.

2.3 Positive versus Normative Economics

Positive economics is the branch of economics analysis that *describes* the way the economy actually works. *Normative economics* make *prescriptions* about the way the economy should work. A *forecast* is a simple prediction of the future.

3 Supply and Demand

Supply and demand analysis helps us understand factors impacting goods' prices and quantity in the economy.

3.1 Competitive Markets

A *competitive market* has **many** buyers and sellers of the **same** good or service, and none of whom can influence the price. Most markets are not perfectly competitive. The *supply and demand model* is a model of how a competitive market behaves.

3.2 Demand

Demand represents the behavior of buyers. The *demand curve* is a plot that shows the quantity demanded of a good at each price level, assuming all other determinants of demand constant. A *demand schedule* is a table showing how much of a good or service consumers will want to buy at different prices. The *quantity demanded* is the quantity that buyers are willing (and able) to purchase at a particular price. The *law of demand* states that a higher price for goods leads people to demand smaller quantity of that good (other things equal).

Remark Any change in price (ΔP) will result in a movement along the demand curve (change in quantity demanded)

Remark Any change in consumer income, consumer taste, or consumer preference will result in a movement of the entire demand curve (change in demand)

A **rightward** shift of the demand curve means an increase in demand. A **leftward** shift of the demand curve means a decrease in demand. There are five factors that **shift the demand curve**:

1. Changes in the prices of related goods or services
2. Changes in income
3. Changes in tastes
4. Changes in expectations
5. Changes in the number of consumers

Two goods are *substitutes* if a decrease in the price of one leads to a decrease in demand for the other (or vice versa); *substitutes usually serve a similar function*. Two goods are *complements* if a decrease in the price of one good leads to an increase in the demand for the other (or vice versa); *complements are usually consumed together*.

The effect of changes in income on demand depends on the nature of the good in question:

- A **normal good**: demand increases when income increases.
- An **inferior good**: demand decreases when income increases.

If consumers have a choice about the timing of a purchase, then they buy according to **expectations**. Buyers adjust current spending in anticipation of the direction of future prices in order to obtain the lowest possible price. As the population of an economy changes, the number of buyers of a particular good also changes, thereby changing its demand.

The **market demand curve** is the horizontal sum of the individual demand curves of all consumers.

3.3 Supply

Supply represents the behavior of sellers. A **supply schedule** shows how much of a good or service would be supplied at different prices. A **supply curve** shows the quantity supplied at various prices. The **quantity supplied** is the quantity that producers are willing and able to sell at a particular price.

A **rightward** shift of the supply curve means an increase in supply. A **leftward** shift of the supply curve means a decrease in supply. Some factors that shift the supply curve are:

1. input prices
2. the prices of related goods or services
3. technology
4. expectations
5. the number of producers

An increase in the price of an input makes production more costly for sellers, and thus supply decreases. A fall in the price of an input makes production less costly for sellers, and thus supply increases.

Inputs used in production have *opportunity costs*. Sellers will choose to use inputs whose profit is the highest. . .

- sellers will supply less of a good if its profitability falls
- there are substitutes and complements in production processes

New, better technology, enables producers to spend less on inputs, yet still produce the same amount of output: supply increases. The expectation of a higher price for a good in the future decreases current supply of the good—if sellers can store the good. Sellers will adjust their current offerings in anticipation of the direction of future prices in order to obtain the highest possible price. As producers enter and exit the market, the overall supply changes:

- *Entry* implies more sellers in the market, increasing supply
- *Exit* implies fewer sellers in the market, decreasing supply.

On the supply side, firms have an incentive to charge the highest price possible. On the demand side, buyers have an incentive to search out the lowest price for the good. In a competitive market with many buyers and sellers, this will lead to one equilibrium price for the good. When $Q_S = Q_d$ at a certain price, the market is in **equilibrium**. That is, the amount consumers would purchase at this price is matched exactly by the amount producers wish to sell. The price at which this takes place is the **equilibrium price**, also referred to as the market-clearing price. The quantity of the good or service bought and sold at that price is the **equilibrium quantity**.

There is a **surplus** of a good when the quantity supplied exceeds the quantity demanded. Surpluses occur when the price is above the equilibrium level. **Surpluses** do not last: sellers will reduce price so they can move goods off the shelves.

There is a **shortage** when the quantity demanded exceeds the quantity supplied. Shortages occur when the price is below its equilibrium level. **Shortages** do not last: sellers will realize they can charge higher prices.

An increase in demand leads to a movement along the supply curve to a higher equilibrium price and higher equilibrium quantity. An increase in supply leads to a movement along the demand curve to a lower equilibrium price and higher equilibrium quantity.

Simultaneous shifts in demand and supply can lead to ambiguous changes to the quantity and price...

- **Supply increases and Demand increases:** Quantity increases, but price change is ambiguous
- **Supply increases and Demand decreases:** Price decreases, but quantity change is ambiguous
- **Supply decreases and Demand increases:** Price increases, but quantity change is ambiguous
- **Supply decreases and Demand decreases:** Quantity decreases, but price change is ambiguous

4 Consumer and Producer Surplus

The difference between your willingness to pay the price is **consumer surplus**. A consumer's willingness to pay for a good is the maximum price at which he

or she would buy that good. **Individual consumer surplus** is the gain to an individual buyer from the purchase of a good; the difference between the price paid and what the buyer is willing to pay. **Total consumer surplus** is the sum of individual consumer surpluses of all buyers in a market. Economists often use the term **consumer surplus** to refer to both individual and total consumer surplus.

Remark Consumer surplus is the area below the demand curve but above the price

Producer surplus is the difference between market price and the price at which firms are willing to supply the product. **Individual producer surplus** is the net gain to an individual seller from selling a good. It is equal to the difference between the price received and the seller's cost (the seller's cost includes monetary costs; it may also include other opportunity cost). **Total producer surplus** is the sum of individual producer surpluses of all the sellers in a market. Economists use the term **producer surplus** to refer to both individual and total producer surplus.

$$\text{Producer surplus} = \text{Price} - \text{Cost}$$

Remark Producer surplus is the area below the price, but above the supply curve.

Total surplus is the sum of the producer and consumers surpluses. Markets are usually efficient; there is no way to make some people better off without making other people worse off. Markets are usually efficient because they maximize total surplus. There are three ways which you might (unsuccessfully) try to increase the total surplus:

1. Reallocate consumption among consumers.
2. Reallocate sales among sellers.
3. Change the quantity traded.

Competitive markets are usually efficient:

1. They allocate consumption of the good to the potential buyers who most value it.
2. They allocate sales to the potential sellers who most value the right to sell the good (e.g. who have the lowest cost).
3. They ensure that all transactions are mutually beneficial. Every consumer who makes a purchase values the good more than every seller who makes a sale.

4. They ensure that no mutually beneficial transactions are missed. Every potential buyer who doesn't make a purchase values the good less than every potential seller who doesn't make a sale.

There are, however, three caveats to efficiency:

1. Although a market may be efficient, it isn't necessarily fair.
2. Markets sometimes fail to deliver efficiency.
3. Even when the market equilibrium maximizes total surplus, this doesn't mean that it results in the best outcomes for every individual consumer or producer.

Property rights are the rights of the owners of valuable items, whether resources or goods, to dispose of those items as they choose. Private property rights create and protect incentives to trade with others and to innovate.

An **economic signal** is any piece of information that helps people make better economic decisions. Prices are the most important signals in a market economy because they convey information about other people's costs and their willingness to pay. If prices are high, we can infer that demand for a given good is potentially high.

5 Price Controls and Quotas

Market prices do not necessarily please buyers or sellers: they may lobby the government to help them by altering the price. **Price controls** are legal restrictions on how high or low a market price may go; there are two main types:

- A **price ceiling** is the maximum price sellers are allowed to charge for a good or service (e.g. rent control): usually set *below* equilibrium
 - Side effects: inefficiently low quantity, inefficient allocation to customers, wasted resources, inefficiently low quality, black markets
 - People expend money, effort, and time to cope with shortages caused by the price ceiling
 - A **black market** is a market in which goods or services are bought and sold illegally—either because they are prohibited or because the equilibrium price is illegal. Black markets encourage disrespect for the law and worsens the position of those who are outside the market.
- A **price floor** is the minimum price buyers are required to pay for a good or service (e.g. minimum wage): usually set *above* equilibrium
 - Side effects: deadweight loss, inefficient allocation of sales among sellers, waste of resources, inefficiently high quality, temptation to break the law by selling below the legal price.

- Price floors allow high-cost firms to operate and prevent low-cost firms from entering the industry.
- Price floors encourage black markets because there are willing sellers (and buyers) at illegal prices, so they are tempted to break the law and trade with each other.

Sometimes governments control quantity instead of price. A **quota** is an upper limit, set by the government, on the quantity of some good that can be bought or sold; also referred to as a **quantity control**. A **quota limit** is the total amount of a good under a limit that can be legally transacted. A **license** is the right, conferred by the government, to supply a good.

The **demand price** is the price of a given quantity at which consumers will demand that quantity. The **supply price** is the price of a given quantity at which producers will supply that quantity. The **wedge, or quote, rent** is the difference between the demand price and the supply price at the quota limit: equal to the market price of the license when the license is traded.

Like price controls, quotas impose losses on society such as deadweight loss and black markets.

6 Elasticity

Price elasticity of demand is the measure of price responsiveness.

$$\text{Elasticity} = \frac{\Delta Q}{P \Delta P}$$

$$\text{Price elasticity of demand} = \frac{\text{Percent change in quantity demanded}}{\text{Percent change in price}}$$

A demand is **elastic** when an increase in price reduces the quantity demanded a lot. A demand is **inelastic** when an increase in price reduces quantity demanded just a little.

The percent change calculation depends on the choice of starting point. To solve this problem, we calculate the price elasticity of demand using the midpoint formula for percentage changes.

$$\text{Percent change in } x = \frac{\text{Change in } x}{\text{Average value of } x} \times 100$$

$$\text{Average value of } x = \frac{(\text{Starting value of } x + \text{Final value of } x)}{2}$$

A good can have a price elasticity as low as zero or as high as infinity.

- If a price elasticity < 1 , the demand curve is **inelastic**.

- If a price elasticity > 1 , the demand curve is **elastic**
- If a price elasticity $= 1$, the demand curve is **unit-elastic**

Total revenue is equal to the price times the quantity sold

$$\text{Total Revenue} = P \times Q$$

When demand is **inelastic**, the price dominates the quantity effect, so an increase in price will cause only a slight reduction in the quantity demanded. In this instance, total revenue will rise when the price rises. When demand is **elastic**, the quantity effect dominates the price effect, so an increase in price will cause a significant reduction in the quantity demanded. In this instance, total revenue will fall when price rises. When demand is unit-elastic, the quantity effect equals the price effect, so an increase in price exactly balances the reduction in the quantity demanded. In this instance, total revenue doesn't change.

6.1 Factors that Determine Price Elasticity of Demand

Some factors that determine the price elasticity of demand are as follows...

1. Whether the good is a **necessity** or a **luxury**
 - For **necessities**, quantity demanded does not change much in response to a change in P .
 - For **luxuries**, quantity demanded is more sensitive to a change in price.
2. The **availability of close substitutes**
 - Fewer substitutes makes it harder for consumers to adjust Q when P changes, so **demand is inelastic**.
 - Many substitutes make it easier for consumers to switch brands when prices change, so **demand is elastic**.
3. The **share of income spent on the good**
 - It feels cheaper when we spend a smaller share of income on the good.
 - It feels more expensive when we spend a greater share of income on the good.
4. **Time elapsed since the price change**
 - Less time to adjust means lower elasticity
 - **Over time** consumers can adjust their behavior by finding substitutes (making demand more elastic).

The ***cross-price elasticity of demand*** measures how sensitive the quantity demanded of good A is to the price of good B.

$$\text{Cross-price elasticity of demand} = \frac{\% \text{ Change in quantity of A demanded}}{\% \text{ Change in price of B}}$$

For **substitutes**, cross-price elasticity of demand is **positive**. For **complements**, cross-price elasticity of demand is **negative**.

The ***income elasticity of demand*** measures how sensitive the quantity demanded of a good is to changes in income.

$$\text{Income elasticity of demand} = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in income}}$$

The income elasticity of demand can be used to distinguish normal from inferior goods. For **normal goods**, income elasticity is **positive**. For **inferior goods**, income elasticity is **negative**.

Remark For **income-elastic goods**, income elasticity is greater than 1. For **income-inelastic goods**, income elasticity is positive but less than 1.

Usually, sellers offer more when prices are higher, but how strong is that relationship?

$$\text{Price elasticity of supply} = \frac{\% \text{ Change in quantity supplied}}{\% \text{ Change in price}}$$

The supply curve is **elastic** if a rise in price increases the quantity supplied a lot. The supply curve is **inelastic** if a rise in price increases the quantity supplied just a little.

7 Taxes

Taxes drive a wedge between the price buyers pay and the price sellers receive. The ***incidence*** of a tax is a measure of who really pays it. The incidence of an excise tax doesn't depend on who officially pays the tax.

Remark The most inelastic side of the market bears the majority of the tax burden.

Increasing the tax rate does not necessarily increase revenue. On one side, the tax increase means that the government raises more revenue for each unit sold, which increases tax revenue. On the other side, it reduces the quantity of sales, which decrease tax revenue.

- If the price elasticities of both supply and demand are low, the tax increase won't reduce the quantity of the good sold very much, so tax revenue will definitely rise

- If the price elasticities are high enough, the tax reduces the quantity sold so much that tax revenue falls
- If the price elasticities are high, the result is less certain and dependent on how high or low the initial tax rate was

The **administrative costs** of a tax are the resources used for its collection, for the method of payment, and for any attempts to evade the tax. Taxes cost society time and effort that could have been used elsewhere.

There are two principles of tax fairness. The **benefits principle** states that those who benefit from public spending should bear the burden of the tax that pays for that spending. The **ability-to-pay principle** states that those with greater ability to pay a tax should pay more. There is usually a trade-off between equity and efficiency: the system can be made more efficient only by making it less fair, and vice versa.

The **tax base** is the measure, such as income or property value, that determines how much tax an individual or firm pays.

- **Income tax** depends on income from wages and investments.
- **Payroll tax** depends on the earnings an employer pays an employee.
- **Sales tax** depends on the value of goods sold.
- **Profits tax** depends on a firm's profit
- **Property tax** depends on the value of property such as a home.
- **Wealth tax** is a tax that depends on an individual's wealth.

The **tax structure** specifies how the tax depends on the tax base. A **progressive tax** takes a larger share of the income of high-income taxpayers than of low-income taxpayers. A **regressive tax** takes a smaller share of the income of high-income taxpayers than that of low-income taxpayers. The **marginal tax rate** is the percentage of an increase in income that is taxed away.

8 Decision Making by Individuals and Firms

Our decisions depend on comparing costs with benefits. Because resources are scarce, the true cost of anything is what you must give up to get it. An **explicit cost** is a cost that requires an outlay of money. An **implicit cost** does not require an outlay of money; it is measured by the value, in dollar terms, of benefits that are forgone.

$$\text{Opportunity cost} = \text{Total explicit cost} + \text{Total implicit cost}$$

Economic profit equals revenue minus the opportunity cost of all resources used.

Capital is the total value of assets owned by an individual or firm—physical assets plus financial assets. The **implicit cost of capital** is the opportunity cost of the use of one's capital; that is, the income earned if the capital had been employed in its next best alternative use. **Sunk cost** is a cost that has already been incurred and is not recoverable; a sunk cost should be ignored in decisions about future actions.

There are two types of decisions: a choice between two alternatives (**either-or**) or a more complex choice that requires us to choose at the margin (**how much**). When faced with an **either-or** choice between two activities, choose the one with the positive economic profit. **How-much** decisions are at the margin. **Marginal analysis** is the comparing the benefit of doing a little bit more of something with the cost of doing a little bit more of something. Must ask if marginal cost is greater than or less than marginal benefit. **Marginal cost** of producing a good or services is the additional cost incurred by producing one more unit of that good or service. **Marginal benefit** is the additional benefit derived from producing one more unit of a good or service.

The **marginal cost curve** shows how the cost of producing one more unit depends on the quantity that has already been produced. Each product has a unique marginal cost, but there are some basic shapes:

- **increasing marginal cost**: each additional unit costs more to produce than the previous one
- **constant marginal cost**: each additional unit costs the same to produce as the previous one
- **decreasing marginal cost**: each additional unit costs less to produce than the previous one

Marginal benefit is the additional benefit derived from producing one more unit of a good or service. There is a **decreasing marginal benefit** from an activity when each additional unit of the activity yields less benefit than the previous unit. The **marginal benefit** curve shows how the benefit from producing one more unit depends on the quantity that has already been produced.

The **optimal quantity** is the quantity that generates the highest possible total profit. **Profit-maximizing principle of marginal analysis** is the largest quantity at which the marginal benefit is greater than or equal to marginal cost.

8.1 Behavioral Economics

We assume that humans are mostly rational; a **rational** decision maker always chooses the available option that leads to the outcome he or she most prefers.

Some reasons that people might rationally choose a worse payoff are as follows:

- **Concerns about fairness**
- **Nonmonetary rewards**
- **Bounded rationality:** making a choice that is close to but not exactly the one that gives you the best payoff
- **Risk aversion:** willing to sacrifice some economic payoff in order to avoid a potential loss

An *irrational* decision maker chooses an option that leaves him or her worse off than choosing another available option that would have left him or her better off. There are seven **common mistakes** in economic decision-making:

1. **Misperceptions of opportunity costs:** people tend to ignore opportunity costs when they are nonmonetary.
2. **Overconfidence:** we tend to think that we know more than we actually do.
3. **Unrealistic expectations about future behavior:** most of us are overly optimistic about our future behavior and level of discipline.
4. **Counting dollars unequally:** the habit of assigning dollars to different accounts so that some dollars are worth more than others (*mental accounting*)
5. **Loss aversion:** an oversensitivity to loss that leads to an unwillingness to recognize a loss and move on.
6. **Framing bias:** the tendency to make a decision based on how the choices are presented
7. **Status quo bias:** the tendency to avoid making a decision altogether.

9 The Rational Consumer

We suppose that each individual is trying to maximize satisfaction gained from consumption. *Utility* is the value or satisfaction from consumption. An individual's *consumption bundle* is the collection of all the goods and services consumed by that individual. An individual's *utility function* gives the total utility generated by his or her consumption bundle. *Util* is a unit of utility. *Marginal utility (MU)* is the change in utility from consuming an additional unit. *Diminishing marginal utility* is the fact that each additional unit of a good adds less to utility than the previous unit.

Because the amount of money a consumer can spend is limited, a decision to consume more of one good is also a decision to consume less of some other

good. A ***budget constraints*** requires that the cost of a consumer's consumption bundle be no more than the consumer's income. A consumer's ***consumer possibilities*** are the set of all consumption bundles that can be consumed given the consumer's income and prevailing prices. A consumer's ***budget line*** shows the consumption bundles available to a consumer who spends all of his or her income.

The ***optimal consumption bundle*** is the one that maximizes a consumer's total utility given his or her budget constraint. We can find optimal consumption choice by choosing the bundle at which total utility is maximized, but we can also use marginal analysis instead. . .

Marginal utility is the additional utility the consumer gets from consuming one more unit of a good or service. The ***marginal utility per dollar*** spent on a good or service is the additional utility from spending one more dollar on that good or service. To find the marginal utility per dollar, we must divide the marginal utility of the good by its price.

Remark Marginal utility per dollar declines as the quantity of that good consumed rises, due to diminishing marginal utility.

The general rule is to compare the marginal utility and the price for all goods, and then adjust your spending toward the goods that give you more marginal utility per dollar. The ***optimal consumption rule*** states that a consumer chooses a consumption bundle for which the marginal utility per dollar spent on all goods is the same.

If the price rises, the marginal utility per dollar spent on the good falls, and the consumer can increase his or her utility by buying less of this good and more of other goods. The ***substitutions effect*** (of a change in the price of a good) is the change in the quantity consumed of that good as the consumer substitutes the good that has become relatively cheaper for the good that becomes relatively more expensive. The ***income effect*** (of a change in the price of a good) is the change in the quantity consumed of a good that results from a change in the consumer's purchasing power due to the change in the price of the good; a change in the price of a good effectively changes a consumer's income. For most goods, the income effect has no significant effect on consumption. Most market demand curves slope downwards because of the substitution effect.

In the case of an inferior good, the income and substitution effects work in opposite directions. A ***giffen good*** is a very rare inferior good for which the income effect outweighs the substitution effect and the demand curve slopes upward.

10 Behind the Supply Curve: Inputs and Costs

A *firm* is an organization that produces goods or services for sale. *Production* is the process of turning inputs into outputs. A *production function* is the relationship between the quantity of inputs a firm uses and the quantity of output it produces. A *fixed input* is an input whose quantity is fixed for a period of time and cannot be varied. A *variable input* is an input whose quantity the firm can vary at any time.

The *long run* is the period in which all inputs can be varied. The *short run* is the period in which at least one input is fixed. The *total production curve* shows how the quantity of output depends on the quantity of the variable input for a given quantity of the fixed input.

The *marginal product* of an input is the additional quantity of output that is produced by using one more unit of that input. *Marginal produce of labor (MPL)* is the change in output resulting from a one-unit increase in the amount of labor input ($\Delta Q/\Delta L$). MPL equals the slope of the total product curve. The MPL is defined as the increase in quantity of output when you increase the quantity of that input by one unit.

A *fixed cost* is a cost that does not depend on the quantity of output produced. It is the cost of the fixed input. A *variable cost* is a cost that depends on the quantity of output produced. It is the cost of the variable input.

The *total cost* of producing a given quantity of output is the sum of the fixed cost and the variable cost of producing that quantity of output.

$$TC = FC + VC$$

The *total cost curve* shows how total cost depends on the quantity of output. The total cost curve becomes steeper as more output is produced, a result of diminishing returns. With diminishing returns, additional units of output require more and more labor; therefore the cost increases. The *marginal cost* is the change in total cost generated by one additional unit of output

$$MC = \Delta TC/\Delta Q$$

where Δ is change, TC is total cost, and Q is quantity of output.

Average total cost is equal to the total cost per unit of output produced

$$ATC = TC/Q$$

Average fixed cost is equal to the fixed cost per unit of output produced

$$AFC = FC/Q$$

Average variable cost is the variable cost per unit of output produced

$$AVC = VC/Q$$

Increasing output has two opposite effects on average total cost:

- **The spreading effect:** the larger the output, the more output over which fixed cost is spread, leading to lower average fixed cost.
- **The diminishing returns effect:** the larger the output, the more variable input required to produce additional units, which leads to higher average variable cost.

Remark At high levels of output, the spreading effect is weaker than the diminishing returns effect

The **minimum-cost output** is the quantity of output at which average total cost is lowest—the bottom of the U-shaped average total cost curve. Three general principles are always true about a firm's marginal cost and average total cost curves:

1. At the minimum-cost output, the average total cost is equal to marginal cost.
2. At output less than the minimum-cost output, marginal cost is less than average total cost and average total cost is falling.
3. At output greater than minimum-cost output, marginal cost is greater than average total cost and average total cost is rising.

Marginal cost curves often slope downward as the output goes from zero to some low level, and they slope upward at higher levels of production. The initial downward slope occurs when employing more workers allows them to specialize in various tasks. This specialization leads to increasing returns to the hiring of additional workers and results in the marginal cost curve sloping downward. Once enough workers exhaust the benefits of specialization, diminishing returns of labor set in and the marginal cost curve slopes upward.

All inputs are variable in the long run. This means that in the long run, fixed cost may also vary. The firm will choose its fixed cost in the long run based on the level of output it expects to produce. There is a trade-off between higher fixed cost and lower variable cost for any given output level, and vice versa.

The **long-run average total cost curve** shows the relationship between output and average total cost when fixed cost has been chosen to minimize average total cost for each level of output. There are **increasing returns to scale** when long-run average total cost declines as output increases. There are **decreasing returns to scale** when long-run average total cost increases as output increases. There are **constant returns to scale** when long-run average total cost is constant as output increases.